Gurinder Singh

Staff Software Engineer (Mobile & AI)

LLM Agents · RAG Pipelines · Mobile · Automation

Portfolio LinkedIn gsingh622@yahoo.com

SCAN FOR PORTFOLIO

90M+	28+	2×	
USERS IMPACTED	HACKATHONS	1ST PLACE WINS	PATENT HOLDER

SUMMARY

Staff Software iOS/Mobile & Applied AI Engineer with over a decade of experience architecting and scaling production consumer platforms and AI-powered systems used by millions. Patent holder and technical leader with deep expertise in iOS platform architecture, distributed client systems, and AI-enabled product development—focused on translating emerging AI capabilities into reliable, real-world products.

Proven track record of taking ideas from early prototypes to globally deployed products, spanning startup innovation through enterprise-grade systems serving 90M+ users, most recently at PayPal/Venmo. Experienced in building agentic AI systems, RAG pipelines, and LLM-powered workflows alongside large-scale mobile features and strategic partnerships, with a strong product mindset and comfort operating in ambiguous, cross-functional environments.

Brings a human-centered perspective to applied AI, emphasizing the elevation of human skill, judgment, and ethos through careful system design. Actively invested in mentoring engineers, shaping technical culture, volunteering with NYC high schools and nonprofits, and building products grounded in context, nuance, and real-world use at scale.

TECHNICAL SKILLS PORTFOLIO

** Condensed overview — full technical depth available on portfolio*

CATEGORY	SPECIALIZED DOMAINS	TOOLS & TECHNOLOGIES	KEY TECH
Mobile	iOS Core Spatial & Hardware Cross-Platform	Swift, Objective-C, SwiftUI, UIKit, SwiftData, Core Data, ARKit, RoomPlan, BLE, NFC, MapKit, CoreML, MLX, RxSwift, Combine, Flutter & Dart	
AI/ML	Models & LLMs Orchestration Media & Vision	Python, PyTorch, OpenAI, Anthropic, Gemini, Llama LangChain, LlamaIndex, CrewAI, N8N, MCP Servers Stable Diffusion, YOLO, OpenCV, Whisper, ElevenLabs	
Cloud & DevOps	Infrastructure BaaS & CI/CD	AWS (Lambda/EC2), GCP (Vertex AI), Azure, Docker Supabase, Firebase, GitHub Actions, Jenkins, Fastlane	
Data & Tools	Databases API & Search Workflow	PostgreSQL, MongoDB, Pinecone, ChromaDB, Redis,GraphQL, Figma, Jira, TDD, Agile, Cursor, Xcode	

PROFESSIONAL EXPERIENCE

PayPal / Venmo

2020 – Present

Staff iOS Engineer → Applied AI/ML Engineer

🤖 AI/ML Evolution & Technical Innovation (2023–Present)

Agent0 — Autonomous AI Agent for Engineering Teams

February 2026

- Deployed Agent Zero (agent0), an open-source agentic AI framework, to Venmo and PayPal engineering teams — enabling autonomous task execution, intelligent tool use, and self-correcting workflows across development pipelines
- Serves as the next evolution of Project Mercury — powering agentic e-commerce experiences for merchants and customers, while also driving internal org-wide workflow automation

Project Mercury — AI-Powered Agentic E-Commerce Platform

June 2025 → Present

- Built fully functional prototype in 72 hours; presented to SVP of AI, earning immediate backing and leading a high-velocity team
- Trained five SDXL LoRA models achieving **23% loss reduction** and 109MB optimized models for production deployment

AI/ML Implementation & Automation (2024)

- Developed AI-powered crash detection system using RAG pipelines + Vector stores with automated DataDog monitors — **now adopted team-wide**
- Created **10%+ productivity improvements** via AI-powered automation tools
- Championed AI coding assistants across teams, training colleagues on leveraging AI

📱 iOS Engineering Excellence & Business Impact

Major Product Launches & Partnerships

- **Pay With Venmo Checkout (2024-Present):** Reduced end-to-end pay sheet render latency by ~2 seconds
- **Venmo Gift Cards (2023):** Led iOS architecture and delivery for Venmo Gift Cards, supporting major merchants like Amazon and Starbucks
- **Venmo × Amazon (2022):** Built sandbox/integration app in two days for partnership launch
- **QRC Touch-Free Payments (2021):** Led moonshot initiative winning **$60M CVS contract**
- **Amex Partnership (2020):** Built payment-splitting integration for **$400M partnership deal**

iOS Innovation & Patents

- **QRC Widget Patent (2022):** Filed "Interface Widget Tool for Automatic QR Code Generation" (PayPal Ref. OCP.D2021.103924.US1). First team to use SwiftUI for iOS widgets at Venmo. **Reduced user friction by 66%**
- **Dynamic Island / Live Activities / App Clip:** Built end-to-end infrastructure including HLDs, LLDs, and APNS notification systems
- **Visual Snapshot Testing:** Built comprehensive visual snapshot testing suite using XCTest for UI regression testing

👥 Technical Leadership & Impact

- Partnered with Design Systems to create modern SwiftUI component library that **cut feature development time by 30%**

Google Stadia

2019 – 2020

iOS Developer/Consultant

- Worked on Google Stadia for mobile platforms using Flutter/Dart and native iOS plugins
- Implemented Core Bluetooth integration for Stadia controller connectivity and low-latency communication
- Utilized Xcode Instruments for performance profiling and CPU/GPU optimization

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Morgan Stanley

2018 – 2019

iOS Developer/Consultant

- Worked on Hybrid (Cordova + Native) Banking and Wealth Management apps for iOS
- Developed key features: check-deposit, biometric authentication, certificate-pinning, jailbreak detection, push authentication
- Migrated codebase from 95% Objective-C to 80% Objective-C / 20% Swift

HACKATHON ACHIEVEMENTS & INNOVATION

Featured highlights – full hackathon history on portfolio

28+ hackathons attended across **6+ countries** as participant, mentor, and judge. Earned **2 first-place wins** and served as judge at Yale (2×). Recognized globally for innovation and technical excellence.

1ST PLACE

MLH @ Bloomberg London (2017)

Automated Search & Rescue System. Featured in BBC documentary.

1ST PLACE

Devcamp NYC (2016)

Talent Management App. Won among 300+ participants at Google NYC.

FINALIST

HackZurich (2018)

AR Car Buying Tool. Finalist among 400+ participants.

Hack Yale – Judge (2×)

Yale University

PenApps – Mentor

University of Pennsylvania

SELECT PERSONAL PROJECTS